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RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences

First Year - Semester II Examination - November/December 2023

MAA 1203 - NUMERICAL ANALYSIS I

Time: Two (02) hours.

- **Answer ALL (04) questions.**
- **Calculators will be provided.**

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1. a) Let $a = 9890.9$ and $b = 9887.1$. Use decimal floating point representation with three significant digits in the mantissa to calculate $a - b$. Do the calculations first by using chopping and then by using rounding. Calculate round-off error in each case. Which rounding method does give a value closer to the exact answer?

(30 marks)

b) The number $\pi = 3.14159265...$ is approximated by $22/7$. Determine up to how many digits is this approximation accurate. Obtain the absolute and relative errors.

(30 marks)

c) Let $x_0, x_1, \cdots, x_{n-1}$ be n points such that $x_{k+1} - x_k = h$(constant) and $y_k = y(x_k)$ for all $k = 0, 1, \ldots, n$. The forward difference operator Δ is defined by $\Delta y_k = y_{k+1} - y_k$.

i. Find $\Delta^2 y_0$, $\Delta^3 y_0$, and $\Delta^4 y_0$.

Contd.

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ii. For the following data for x and y variables, calculate $\Delta^4 y_k$.

x	-4	-2	0	2	4	6
y	-139	-21	1	23	141	451

(40 marks)

2. a) Discuss advantages and disadvantages of the Newton-Raphson method for solving non-linear equations in comparison with the bisection method.

(15 marks)

b) Determining the square root of a number N is equivalent to finding a solution to the equation: $f(x) = x^2 - N = 0$.

i. Derive the Newton-Raphson iteration formula with the starting value x_0 for the n-th iteration, x_n, $n = 1, 2, 3, \cdots$, for the above problem.

ii. Perform five (5) iterations of the Newton-Raphson method, using the initial approximation $x_0 = 2$, to approximate $\sqrt{2}$. If the exact value for $\sqrt{2}$ is 1.414213562373095,..., find the absolute and relative errors.

(85 marks)

3. Differentiate the difference between the Regula-falsi method and the Secant method for root-finding of nonlinear equations.

(10 marks)

Let $f(x) = x^3 - 5x + 1$.

a) Show that a root (r_0) of $f(x) = 0$ exists between 0 and 1.

(20 marks)

b) Write down the iteration formula of the Regula-falsi method to determine r_0.

(20 marks)

c) Perform four iterations of the Regula-falsi method to obtain r_0.

(50 marks)

4. Briefly, explain disadvantages of Lagrange interpolation method over Newton's interpolation method.

(10 marks)

Consider the following data for x and y variables:

x	1	2.2	3.4
y	2	2.8	3

a) Determine the second-order polynomial in Lagrange form that passes through the points. Use it to calculate the interpolated value of y at 1.8.

(45 marks)

b) Determine the second-order polynomial in Newton's form that passes through the points. Calculate the coefficients of the polynomial using a divided difference table. Use it to calculate the interpolated value of y at 1.8.

(45 marks)

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